Griffiths Problem 7.38: Electron–Muon Scattering

Spin-Averaged Amplitude & Cross Section at High Energy

Including Mandelstam Variable Form of $|\mathcal{M}|^2$

PHYS 7502 — HW Solution

Problem Statement (Griffiths 7.38)

(a) Starting with Equation 7.119 (Casimir's trick result, Eq. 7.126), find the spin-averaged amplitude $\langle|\mathcal{M}|^2\rangle$ for electron–muon scattering ($e^-\mu^- \to e^-\mu^-$) in the CM frame, in the high-energy regime ($m, M \to 0$).

(b) Find the CM differential cross section for electron–muon scattering at high energy. Let E be the electron energy and θ the scattering angle.

What This Problem Wants

- **Part (a):** Start from the spin-averaged $|\mathcal{M}|^2$ (Griffiths Eq. 7.129) and evaluate all dot products in the CM frame with $m = M = 0$. Express the result in terms of E and θ.
- **Mandelstam form (HW addition):** Express $\langle|\mathcal{M}|^2\rangle$ in Lorentz-invariant Mandelstam variables s, t, u. (Cross section not needed for this part.)
- **Part (b):** Plug $\langle|\mathcal{M}|^2\rangle$ into the 2-body CM cross section formula to get $d\sigma/d\Omega$.

1 Setup and Conventions

Griffiths Labeling Convention

For $e^-\mu^- \to e^-\mu^-$, Griffiths assigns:

$$\underbrace{e^-(p_1)}_{\text{in}} + \underbrace{\mu^-(p_2)}_{\text{in}} \;\to\; \underbrace{e^-(p_3)}_{\text{out}} + \underbrace{\mu^-(p_4)}_{\text{out}}$$

There is a **single** t-channel Feynman diagram (virtual photon exchanged between e^- and μ^-). The amplitude is (Griffiths Eq. 7.106):

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]$$

where $g_e^2 = 4\pi\alpha$ (with $\hbar = c = 1$).

2 Spin-Averaged $|\mathcal{M}|^2$ — General Expression

Squaring $|\mathcal{M}|^2$, summing over final spins, averaging over initial spins (factor of $\frac{1}{4}$), and applying Casimir's trick twice gives (Griffiths Eq. 7.126):

$$\langle|\mathcal{M}|^2\rangle = \frac{g_e^4}{4(p_1-p_3)^4}\,\mathrm{Tr}\big[\gamma^\mu(\not{p}_3+mc)\gamma^\nu(\not{p}_1+mc)\big]\,\mathrm{Tr}\big[\gamma_\mu(\not{p}_4+Mc)\gamma_\nu(\not{p}_2+Mc)\big] \tag{1}$$

where m is the electron mass and M is the muon mass. The factor $\frac{1}{4}$ accounts for averaging over the 2 initial electron spins $\times$ 2 initial muon spins.

2.1 Evaluating the Traces

Using trace theorem 13′: $\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = 4[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b\cdot c)]$, and the fact that traces of an odd number of γ-matrices vanish (theorem 10), each trace evaluates to:

$$\begin{aligned}\mathrm{Tr}\big[\gamma^\mu(\not{p}_3+mc)\gamma^\nu(\not{p}_1+mc)\big] &= \mathrm{Tr}\big[\gamma^\mu\not{p}_3\gamma^\nu\not{p}_1\big] + m^2c^2\,\mathrm{Tr}\big[\gamma^\mu\gamma^\nu\big] \\ &= 4\big[p_3^\mu p_1^\nu + p_3^\nu p_1^\mu - g^{\mu\nu}(p_1\cdot p_3) + m^2c^2\,g^{\mu\nu}\big]\end{aligned} \tag{2}$$

Similarly:

$$\mathrm{Tr}\big[\gamma_\mu(\not{p}_4+Mc)\gamma_\nu(\not{p}_2+Mc)\big] = 4\big[p_{4\mu}p_{2\nu} + p_{4\nu}p_{2\mu} - g_{\mu\nu}(p_2\cdot p_4) + M^2c^2\,g_{\mu\nu}\big] \tag{3}$$

2.2 Contracting the Two Traces

We must contract (2) with (3) over μ and ν. The full result (Griffiths Eq. 7.129) is:

General Result (Griffiths Eq. 7.129)

$$\begin{aligned}\langle|\mathcal{M}|^2\rangle = \frac{8g_e^4}{(p_1-p_3)^4}\big[&(p_1\cdot p_2)(p_3\cdot p_4) + (p_1\cdot p_4)(p_2\cdot p_3) \\ &- (p_1\cdot p_3)(Mc)^2 - (p_2\cdot p_4)(mc)^2 + 2(mMc^2)^2\big]\end{aligned} \tag{4}$$

The contraction works as follows (writing $L^{\mu\nu} \equiv p_3^\mu p_1^\nu + p_3^\nu p_1^\mu + g^{\mu\nu}(m^2c^2 - p_1\cdot p_3)$ and similarly for the muon tensor):

- $p_3^\mu p_1^\nu \cdot p_{4\mu}p_{2\nu} = (p_3\cdot p_4)(p_1\cdot p_2)$
- $p_3^\mu p_1^\nu \cdot p_{4\nu}p_{2\mu} = (p_3\cdot p_2)(p_1\cdot p_4)$
- $p_3^\mu p_1^\nu \cdot (-g_{\mu\nu})(p_2\cdot p_4 - M^2c^2) = -(p_1\cdot p_3)(p_2\cdot p_4 - M^2c^2)$
- $p_3^\nu p_1^\mu \cdot p_{4\mu}p_{2\nu} = (p_1\cdot p_4)(p_3\cdot p_2)$
- $p_3^\nu p_1^\mu \cdot p_{4\nu}p_{2\mu} = (p_3\cdot p_4)(p_1\cdot p_2)$
- $p_3^\nu p_1^\mu \cdot (-g_{\mu\nu})(p_2\cdot p_4 - M^2c^2) = -(p_1\cdot p_3)(p_2\cdot p_4 - M^2c^2)$
- $g^{\mu\nu}(m^2c^2 - p_1\cdot p_3)\cdot p_{4\mu}p_{2\nu} = (p_2\cdot p_4)(m^2c^2 - p_1\cdot p_3)$
- $g^{\mu\nu}(m^2c^2 - p_1\cdot p_3)\cdot p_{4\nu}p_{2\mu} = (p_2\cdot p_4)(m^2c^2 - p_1\cdot p_3)$
- $g^{\mu\nu}(m^2c^2 - p_1\cdot p_3)\cdot(-g_{\mu\nu})(p_2\cdot p_4 - M^2c^2) = -4(m^2c^2 - p_1\cdot p_3)(p_2\cdot p_4 - M^2c^2)$

Collecting the $16\times$ factor from the two traces into the prefactor (giving $16/4 = 4$ times the $g_e^4/(p_1-p_3)^4$ prefactor, hence $4g_e^4/\ldots$, times a factor of 2 from grouping symmetric terms), one arrives at (4).

3 Part (a): High-Energy Limit $m = M = 0$

Setting $m = M = 0$, Eq. (4) simplifies to:

$$\langle|\mathcal{M}|^2\rangle = \frac{8g_e^4}{(p_1 - p_3)^4}\left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\right] \tag{5}$$

3.1 CM Frame Kinematics ($m = M = 0$)

In the CM frame with all masses set to zero, each particle has energy E and $|\mathbf{p}| = E$ (since $E^2 = p^2 + m^2 \to E = |\mathbf{p}|$). The four-momenta are:

$$\begin{aligned} p_1 &= (E, \mathbf{p}) & p_2 &= (E, -\mathbf{p}) \\ p_3 &= (E, \mathbf{p}') & p_4 &= (E, -\mathbf{p}') \end{aligned} \tag{6}$$

where $|\mathbf{p}| = |\mathbf{p}'| = E$ and $\mathbf{p} \cdot \mathbf{p}' = E^2 \cos\theta$.

3.2 Evaluating the Dot Products

$$p_1 \cdot p_2 = E \cdot E - \mathbf{p} \cdot (-\mathbf{p}) = E^2 + E^2 = 2E^2 \tag{7}$$

$$p_3 \cdot p_4 = E \cdot E - \mathbf{p}' \cdot (-\mathbf{p}') = 2E^2 \tag{8}$$

$$p_1 \cdot p_3 = E^2 - \mathbf{p} \cdot \mathbf{p}' = E^2 - E^2 \cos\theta = E^2(1 - \cos\theta) \tag{9}$$

$$p_1 \cdot p_4 = E^2 - \mathbf{p} \cdot (-\mathbf{p}') = E^2 + E^2 \cos\theta = E^2(1 + \cos\theta) \tag{10}$$

$$p_2 \cdot p_3 = E^2 - (-\mathbf{p}) \cdot \mathbf{p}' = E^2 + E^2 \cos\theta = E^2(1 + \cos\theta) \tag{11}$$

Useful identities

$1 - \cos\theta = 2\sin^2(\theta/2)$ and $1 + \cos\theta = 2\cos^2(\theta/2)$.

3.3 The Propagator Denominator

$$(p_1 - p_3)^2 = p_1^2 + p_3^2 - 2p_1 \cdot p_3 = 0 + 0 - 2E^2(1 - \cos\theta) = -4E^2 \sin^2(\theta/2) \tag{12}$$

So:

$$(p_1 - p_3)^4 = 16E^4 \sin^4(\theta/2) \tag{13}$$

3.4 Assembling the Numerator

$$(p_1 \cdot p_2)(p_3 \cdot p_4) = (2E^2)(2E^2) = 4E^4 \tag{14}$$

$$(p_1 \cdot p_4)(p_2 \cdot p_3) = E^2(1 + \cos\theta) \cdot E^2(1 + \cos\theta) = E^4(1 + \cos\theta)^2 = 4E^4 \cos^4(\theta/2) \tag{15}$$

3.5 Final Result for Part (a)

$$\begin{aligned}\langle|\mathcal{M}|^2\rangle &= \frac{8g_e^4}{16E^4\sin^4(\theta/2)}\left[4E^4 + 4E^4\cos^4(\theta/2)\right] \\ &= \frac{8g_e^4 \cdot 4E^4}{16E^4\sin^4(\theta/2)}\left[1+\cos^4(\theta/2)\right] \\ &= \frac{2g_e^4}{\sin^4(\theta/2)}\left[1+\cos^4(\theta/2)\right] \qquad (16)\end{aligned}$$

Result: Part (a)

$$\boxed{\langle|\mathcal{M}|^2\rangle = \frac{2g_e^4}{\sin^4(\theta/2)}\left[1+\cos^4\left(\frac{\theta}{2}\right)\right]} \qquad (17)$$

where $g_e^2 = 4\pi\alpha$.

4 Mandelstam Variable Form of $\langle|\mathcal{M}|^2\rangle$

Mandelstam Variables — Definition

For $2 \to 2$ scattering with $p_1 + p_2 = p_3 + p_4$:

$$s \equiv (p_1+p_2)^2 = (p_3+p_4)^2 \qquad (18)$$
$$t \equiv (p_1-p_3)^2 = (p_2-p_4)^2 \qquad (19)$$
$$u \equiv (p_1-p_4)^2 = (p_2-p_3)^2 \qquad (20)$$

In the massless limit: $s + t + u = 0$.

4.1 Expressing Dot Products in Terms of s, t, u

With all masses zero:

$$s = (p_1+p_2)^2 = 2\,p_1\cdot p_2 \quad\Longrightarrow\quad p_1\cdot p_2 = \frac{s}{2} \qquad (21)$$
$$t = (p_1-p_3)^2 = -2\,p_1\cdot p_3 \quad\Longrightarrow\quad p_1\cdot p_3 = -\frac{t}{2} \qquad (22)$$
$$u = (p_1-p_4)^2 = -2\,p_1\cdot p_4 \quad\Longrightarrow\quad p_1\cdot p_4 = -\frac{u}{2} \qquad (23)$$

Key step: relating the remaining dot products

Using $s + t + u = 0$ and 4-momentum conservation, the remaining dot products are:

$$p_3 \cdot p_4 = \frac{s}{2} \qquad (\text{since } (p_3 + p_4)^2 = s) \tag{24}$$

$$p_2 \cdot p_3 = -\frac{u}{2} \qquad (\text{since } (p_2 - p_3)^2 = u = -2p_2 \cdot p_3) \tag{25}$$

$$p_2 \cdot p_4 = -\frac{t}{2} \qquad (\text{since } (p_2 - p_4)^2 = t = -2p_2 \cdot p_4) \tag{26}$$

4.2 Substitution into $\langle|\mathcal{M}|^2\rangle$

Starting from Eq. (5):

$$\langle|\mathcal{M}|^2\rangle = \frac{8g_e^4}{t^2}\left[\frac{s}{2}\cdot\frac{s}{2} + \left(-\frac{u}{2}\right)\left(-\frac{u}{2}\right)\right] = \frac{8g_e^4}{t^2}\cdot\frac{s^2+u^2}{4}$$

Result: Mandelstam Form

$$\boxed{\langle|\mathcal{M}|^2\rangle = \frac{2g_e^4(s^2+u^2)}{t^2} = \frac{2(4\pi\alpha)^2(s^2+u^2)}{t^2} = \frac{32\pi^2\alpha^2(s^2+u^2)}{t^2}} \tag{27}$$

This is Lorentz invariant and valid in **any** frame.

4.3 Consistency Check: Recovering the CM-Frame Result

In the CM frame with massless particles:

$$s = 4E^2 \tag{28}$$

$$t = -2E^2(1-\cos\theta) = -4E^2\sin^2(\theta/2) \tag{29}$$

$$u = -2E^2(1+\cos\theta) = -4E^2\cos^2(\theta/2) \tag{30}$$

Substituting:

$$s^2 + u^2 = 16E^4 + 16E^4\cos^4(\theta/2) = 16E^4\left[1+\cos^4(\theta/2)\right] \tag{31}$$

$$t^2 = 16E^4\sin^4(\theta/2) \tag{32}$$

$$\langle|\mathcal{M}|^2\rangle = \frac{2g_e^4\cdot 16E^4[1+\cos^4(\theta/2)]}{16E^4\sin^4(\theta/2)} = \frac{2g_e^4[1+\cos^4(\theta/2)]}{\sin^4(\theta/2)}\ \checkmark$$

This matches Eq. (17) exactly.

5 Part (b): CM Differential Cross Section

5.1 Cross Section Formula

For 2-body $\to$ 2-body scattering in the CM frame (Griffiths Eq. 6.47), when initial and final particles are massless:

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{\langle|\mathcal{M}|^2\rangle}{(E_{\text{CM}})^2} \tag{33}$$

where $E_{\text{CM}} = E_1 + E_2 = 2E$ is the total CM energy (so $E_{\text{CM}}^2 = s = 4E^2$). The factor of $|\mathbf{p}_f|/|\mathbf{p}_i| = 1$ since $|\mathbf{p}_i| = |\mathbf{p}_f| = E$ in the massless limit. In natural units ($\hbar = c = 1$):

$$\frac{d\sigma}{d\Omega} = \frac{\langle|\mathcal{M}|^2\rangle}{64\pi^2 s} \tag{34}$$

5.2 Plugging In

Using the CM result (17) with $g_e^2 = 4\pi\alpha$ and $s = 4E^2$:

$$\begin{aligned}\frac{d\sigma}{d\Omega} &= \frac{1}{64\pi^2 \cdot 4E^2} \cdot \frac{2(4\pi\alpha)^2}{\sin^4(\theta/2)}\left[1 + \cos^4(\theta/2)\right] \\ &= \frac{2 \cdot 16\pi^2\alpha^2}{256\pi^2 E^2 \sin^4(\theta/2)}\left[1 + \cos^4(\theta/2)\right] \\ &= \frac{\alpha^2}{8E^2 \sin^4(\theta/2)}\left[1 + \cos^4(\theta/2)\right] \end{aligned} \tag{35}$$

Or equivalently:

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2E}\right)^2 \frac{1}{2\sin^4(\theta/2)}\left[1 + \cos^4(\theta/2)\right] \tag{36}$$

Result: Part (b)

$$\boxed{\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2E}\right)^2 \frac{1 + \cos^4(\theta/2)}{2\sin^4(\theta/2)}} \tag{37}$$

This matches the answer given in Griffiths.

5.3 Lorentz-Invariant Form of the Cross Section

We can also write the cross section using Mandelstam variables. From Eq. (27):

$$\frac{d\sigma}{d\Omega} = \frac{2g_e^4(s^2 + u^2)}{64\pi^2 s\, t^2} = \frac{\alpha^2(s^2 + u^2)}{2s\, t^2} \tag{38}$$

Or, converting to $d\sigma/dt$ using $dt = -\frac{s}{2}\, d(\cos\theta)$ and $d\Omega = 2\pi\, d(\cos\theta)$, so $d\Omega = -\frac{4\pi}{s}\, dt$:

$$\frac{d\sigma}{dt} = \frac{s}{4\pi} \cdot \frac{d\sigma}{d\Omega}\bigg/ s = \frac{2\pi\alpha^2(s^2 + u^2)}{s^2 t^2} \tag{39}$$

6 Physical Discussion

Physical Insight

1. **Forward singularity:** The $\sin^{-4}(\theta/2)$ factor (or $1/t^2$ in Mandelstam language) produces a strong forward peak. This is the hallmark of t-channel photon exchange: the virtual photon propagator $\sim 1/t$ diverges as $t \to 0$ ($\theta \to 0$). Physically, small-angle scattering corresponds to large impact parameters where the Coulomb potential is weak but the phase space is large.

2. **Structure of the numerator:** The $s^2 + u^2$ factor encodes the spin structure. The s^2 term comes from $(p_1 \cdot p_2)(p_3 \cdot p_4)$, which is helicity-conserving. The u^2 term comes from $(p_1 \cdot p_4)(p_2 \cdot p_3)$, corresponding to helicity-flip contributions. In terms of CM angles: $s^2 + u^2 \propto 1 + \cos^4(\theta/2)$.

3. **Comparison with Rutherford:** At $\theta \to 0$ the $\cos^4(\theta/2) \to 1$ term doubles the cross section compared to a pure $1/\sin^4(\theta/2)$ Rutherford form. The deviation from Rutherford at large angles is a relativistic spin effect.

4. **Crossing to $e^+e^- \to \mu^+\mu^-$:** Under the crossing $s \leftrightarrow t$, the Mandelstam form $2g_e^4(s^2 + u^2)/t^2$ for t-channel $e\mu$ scattering becomes $2g_e^4(t^2 + u^2)/s^2$ for s-channel $e^+e^- \to \mu^+\mu^-$ annihilation (cf. Thomson Eq. 6.26).

7 Summary of Results

Quantity	**Expression** ($m = M = 0$)
$\langle\lvert\mathcal{M}\rvert^2\rangle$ (CM frame)	$\dfrac{2g_e^4}{\sin^4(\theta/2)}\left[1 + \cos^4(\theta/2)\right]$
$\langle\lvert\mathcal{M}\rvert^2\rangle$ (Mandelstam)	$\dfrac{2g_e^4(s^2 + u^2)}{t^2}$
$d\sigma/d\Omega$ (CM frame)	$\left(\dfrac{\alpha}{2E}\right)^2 \dfrac{1 + \cos^4(\theta/2)}{2\sin^4(\theta/2)}$
$d\sigma/d\Omega$ (Mandelstam)	$\dfrac{\alpha^2(s^2 + u^2)}{2s\,t^2}$